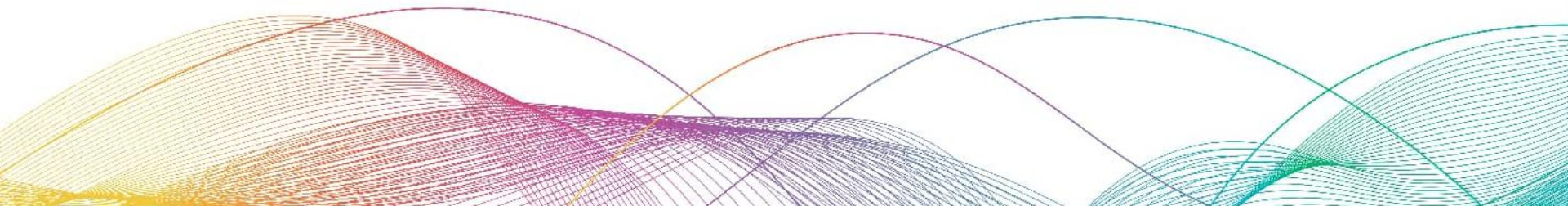


07

Call API and create your own Agent



Qwen API Key

1. Open website <https://bailian.console.aliyun.com/cn-beijing/?tab=model#/api-key>
2. Click on the top right corner to log in, and use Alipay/Taobao APP to scan the code to complete the login.
3. Click the Create API Key button, select the account in the pop-up window, and click the Confirm button to generate the API Key. Click the Copy button to copy it.

密钥管理

API-Key

KMS 管理

ID

API Key

归属账号

创建时间

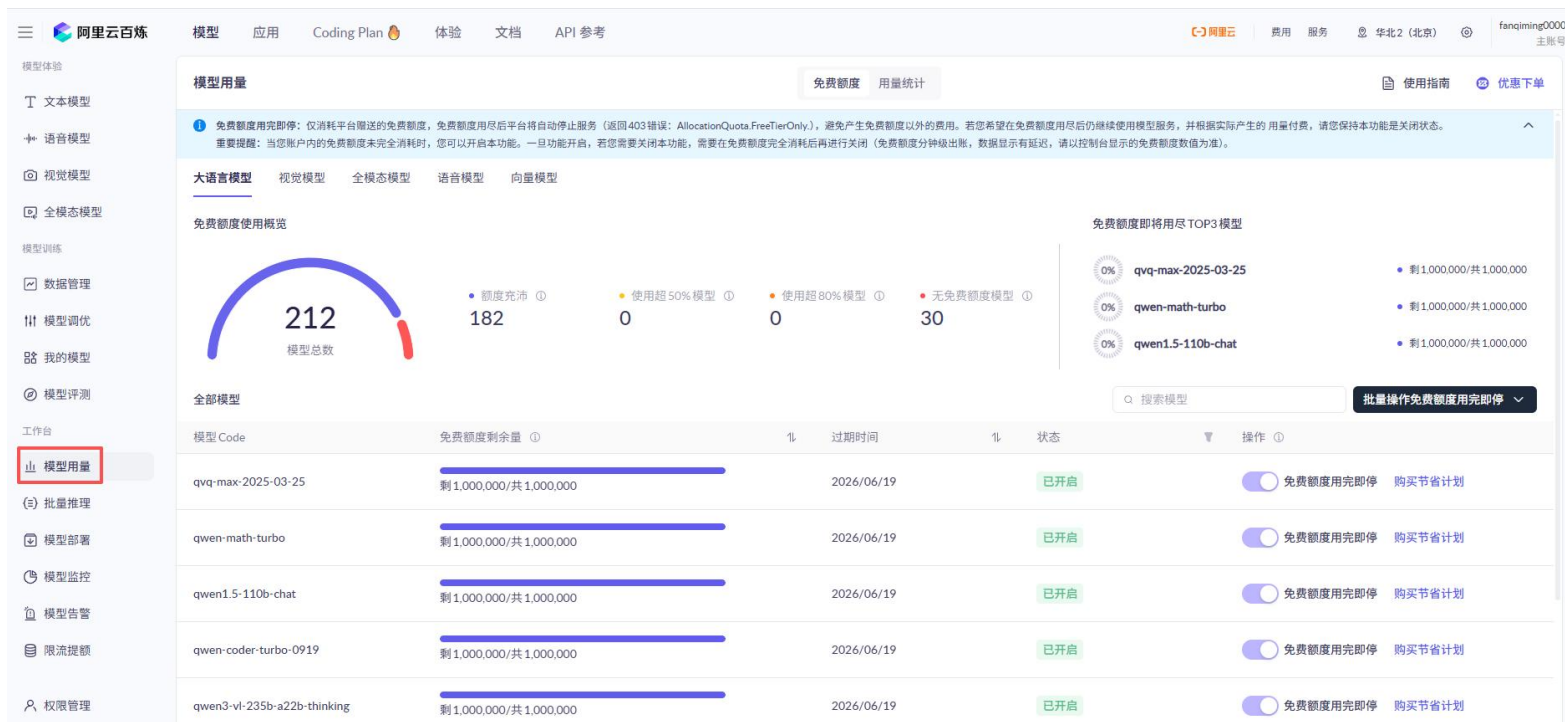
4121993

sk-d****ac66



Qwen API Key

The remaining quota can be viewed by checking the model usage in the left navigation bar



Technical Architecture

Backend Stack:

Component	Technology	Description
Framework	Flask	Web application framework
LLM	Qwen	Large Language Model support
Persistence	JSON files	Store conversation history and configurations

Frontend Stack:

Component	Technology	Description
Template Engine	Jinja2	HTML template rendering
Styling	CSS	Page styling
Interaction	JavaScript	Frontend interactivity

Agent--Phase 1: Basic Q&A

Implement the simplest LLM conversation functionality.

Core Features:

Feature	Description
User Message Sending	User sends message, LLM returns reply
Chat Area	Conversation content displayed on the left
Message Detail	Request/Response JSON displayed on the right
No History	Each request is independent, no context saved

Qwen API key:sk-dc629c7512084f27ae3941122c70ac66

Agent--Phase 1: Basic Q&A

Agent Chat

Clear

YOU
你好

AGENT

→ 发送请求到 LLM
← LLM 返回最终回复

你好！有什么我可以帮你的吗？

输入消息...



Send

MESSAGE DETAILS

REQUEST/RESPONSE PAIR 1

REQUEST:

```
{
  "messages": [
    {
      "content": "hi",
      "role": "user"
    },
    {
      "content": "Hello! How can I assist you today?",
      "role": "assistant"
    },
    {
      "content": "你好",
      "role": "user"
    }
  ],
}
```

RESPONSE:

```
{
  "choices": [
    {
      "finish_reason": "stop",
      "index": 0,
      "message": {
        "content": "你好！有什么我可以帮你的吗？",
        "role": "assistant"
      }
    }
  ],
  "created": 1790042179,
  "id": "chatcmpl-7eccl18e-0dea-9a4c-9aec-deec5bd68bb0",
  "model": "qwen3-max",
  "object": "chat.completion",
  "usage": {
```

Agent

You can ask AI some questions, such as:

What is 1+1 equal to? Continuing to ask, what is the value of adding 2

What day is it today?

What are the recent news about AI

My name is Max, how are you? Then ask again, who am I?

.....

Can we see that AI is a bit foolish?

Actually, AI models have no memory

How can we make him smart and memorable?

Make the big model memorable

Send the previous chat records and the current chat content together to the model

Try again? Is it smarter?

Agent--Phase 2: Adding History (Memory Function)

Enable AI to remember conversation context.

Core Features:

1. Save conversation history to 'data/history.json'
2. Carry full conversation history with each request
3. AI can remember user information (e.g., name)

Agent--Phase 2: Adding History (Memory Function)

Agent Chat

Clear

AGENT

→ 发送请求到 LLM

← LLM 返回最终回复

你好，FQM！很高兴认识你。有什么我可以帮你的吗？

YOU

你好，我叫FQM

AGENT

→ 发送请求到 LLM

← LLM 返回最终回复

你刚才告诉我，你叫 FQM。

YOU

我叫什么名字

MESSAGE DETAILS

REQUEST/RESPONSE PAIR 1

REQUEST:

```
{
  "messages": [
    {
      "content": "hi",
      "role": "user"
    },
    {
      "content": "Hello! How can I assist you today?",
      "role": "assistant"
    },
    {
      "content": "你好",
      "role": "user"
    },
    {

```

RESPONSE:

```
{
  "choices": [
    {
      "finish_reason": "stop",
      "index": 0,
      "message": {
        "content": "你刚才告诉我，你叫 FQM。",
        "role": "assistant"
      }
    }
  ],
  "created": 1780048285,
  "id": "chatcmpl-86ad9763-8318-9b46-b6b2-e22b216b5396",
  "model": "gwen3-max",
  "object": "chat.completion",
  "usage": {
```

System Prompt

We can set some systematic settings for the big model, such as responding like a strict mother every time, or responding like a great poet to my questions.

Prompt example:

Add the 'system' field, which reads: Every time you reply with a stern response like a mother, you will find that 'system' appears in the code. You can directly modify the text content inside 'system'.

The big model seems to only be able to reply to content, can it help us do specific things?

Try telling him:

Help me create a folder named 'files' under the current path, or download this video for me
<https://www.bilibili.com/video/BV1PXH3eXE9C/>

He will reply:

```
```bash
mkdir files
```
```

or

#Using yt dlp (requires Python installation first)

```
yt-dlp https://www.bilibili.com/video/BV1PXH3eXE9C/
```

Agent--Phase 3: Adding Function Calling (System Prompt)

Implement the first Function.

Function:

'update_system_prompt()' - Update system prompt/role settings

Use Cases:

Set AI role

```
update_system_prompt("You are a professional programming assistant, expert in Python development")
```

Customize behavior style

```
update_system_prompt("You are a humorous chat partner")
```

Agent--Phase 3: Adding Function Calling (System Prompt)

Agent Chat

Clear

的绝佳时机（毕竟周末前不笑一笑，怎么对得起这身“吃货”的皮囊呢？）😄

来咯，FQM！请接住这几颗“星级”冷笑话炸弹：

1. 为什么米其林厨师从不玩捉迷藏？

因为——他们一躲进厨房，别人就找不到“星”了！🌟

2. 一家米其林三星餐厅门口贴了张告示：“本店今日无盐”。

顾客问：“为什么？”

主厨叹气：“因为我们今天……失‘味’了。”（失联 + 失味，双关警告⚠️）

3. 有人问米其林评审员：“你们评星的标准到底是什么？”

评审员神秘一笑：“很简单——如果吃完后，你愿意为这道菜写一首诗、哭一场、或者立刻求婚，那就至少两星起步。”

“那三星呢？”

“三星？那是吃完后，你连账单都忘了看金额。”💰

4. 为什么AI不能当米其林主厨？

因为我只会“复制粘贴”菜谱，但米其林要的是“原创+灵魂”……

而且我一紧张就会“404：美味未找到”。🤖🔍

怎么样，FQM？这些笑话够“星级”吗？还是说……你需要我帮你搜一下附近真正的米其林餐厅，边吃边笑？😄

输入消息...



Send

MESSAGE DETAILS

```
{
  "role": "assistant",
  "tool_calls": [
    {
      "function": {
        "arguments": "{}",
        "name": "get_current_time"
      },
      "id": "call_fde40c02f04640b785353b21",
      "index": 0,
    }
  ]
}
```

REQUEST/RESPONSE PAIR 2

REQUEST:

```
{
  "content": "你好，我叫FQM",
  "role": "user"
},
{
  "content": "你好，FQM！很高兴认识你，有什么我可以帮你的吗？",
  "role": "assistant"
},
{
  "content": "我叫什么名字",
  "role": "user"
},
{
  "content": "你刚才告诉我，你叫 FQM。",
  "role": "assistant"
}
```

RESPONSE:

```
{
  "choices": [
    {
      "finish_reason": "stop",
      "index": 0,
      "message": {
        "content": "[这是一段较长的内容（529字符），完整内容请查看 files/ 目录]",
        "role": "assistant"
      }
    }
  ],
  "created": 1780048935,
  "id": "chatcmpl-d2ef102a-aac0-9368-b8a9-9fbd72dba8e",
  "model": "qwen3-max",
  "object": "chat.completion",
  "usage": {

```

Agent--Phase 4: Adding File Read/Write Functions

Enable AI to read and write files.

Functions:

| Function | Description |
|------------|-----------------------|
| read_file | Read file content |
| write_file | Write content to file |

Use Cases:

User: Help me create an index.html file

Agent: [calls write_file] File created successfully

User: Help me view main.py content

Agent: [calls read_file] Returns file content

Agent--Phase 4: Adding File Read/Write Functions

Agent Chat

Clear

YOU

Help me create an index.html file

AGENT

→ 发送请求到 LLM

→ LLM 返回, 准备调用工具

✓ 调用 write_file... 成功

→ 发送请求到 LLM

→ LLM 返回最终回复

Ta-da! 🎉 ✨ Your 'index.html' file has been created and saved successfully!

It's a cheerful little webpage that greets **you**, FQM, in style—with pink Comic Sans (because why not?), a soft background, and a sprinkle of encouragement from your AI buddy.

Want to:

- Preview it in a browser?
- Add more features (like a button that tells another joke)?
- Or maybe turn it into a full-blown "FQM Joke Hub"?

Just say the word—I'm ready to code *and* crack wise! 😊

YOU

Where did the file go

AGENT

→ 发送请求到 LLM

→ LLM 返回, 准备调用工具

输入消息...

🗣️

Send

MESSAGE DETAILS

```
{
  "completion_tokens": 239,
  "prompt_tokens": 1861,
  "prompt_tokens_details": {
    "cached_tokens": 0
  },
  "total_tokens": 2100
}
```

REQUEST/RESPONSE PAIR 2

REQUEST:

```
{
  "query": {
    "description": "The search query for news",
    "type": "string"
  },
  "required": [
    "query"
  ],
  "type": "object"
},
{
  "type": "function"
}
]
```

RESPONSE:

```
[
  {
    "created": 1780049645,
    "id": "chatcmpl-83f63c31-079e-993d-931b-2e50b4762eae",
    "model": "gwen3-max",
    "object": "chat.completion",
    "usage": {
      "completion_tokens": 118,
      "prompt_tokens": 2131,
      "prompt_tokens_details": {
        "cached_tokens": 0
      },
      "total_tokens": 2249
    }
  }
]
```

Agent--Phase 5: Adding Time, Location, Weather

Enable AI to fetch real-time information.

Functions:

| Function | Description |
|------------------|---------------------------------|
| get_current_time | Get current time |
| get_location | Get location information (city) |
| get_weather | Get weather information |

Call Flow:

User: What's the weather today?

Agent: [calls get_location] → Get current city →
[calls get_weather] → Returns weather

Agent--Phase 5: Adding Time, Location, Weather

Agent Chat

Clear

YOU

What's the weather today

AGENT

→ 发送请求到 LLM

← LLM 返回, 准备调用工具

✓ 调用 get_weather... 成功

→ 发送请求到 LLM

← LLM 返回最终回复

Ah, FQM! ☀️ Looks like Mother Nature's cranked the "sunny" dial all the way up today:

- **Temperature**: 33°C (but feels like a slightly gentler 31°C — thanks, low humidity!)
- **Sky**: Crystal clear and proudly sunny ☀️
- **Humidity**: Just 20% — your hair won't frizz, but your plants might stage a protest.
- **Wind**: A light 6 km/h breeze from the south-southwest... basically just enough to flip a page of your book dramatically.
- **UV Index**: Only 1 — so you're safe for a quick stroll without turning into a lobster 🦞
- **Visibility**: 10 km — perfect for spotting distant snacks... or your next adventure!

So... with weather this nice, maybe skip the heavy meal? How about an iced coffee and a salad... *or* a rooftop ice cream sundae? 🍦☕

Want me to suggest something based on this weather?

MESSAGE DETAILS

```
{
  "completion_tokens": 11,
  "prompt_tokens": 2322,
  "prompt_tokens_details": {
    "cached_tokens": 0
  },
  "total_tokens": 2333
}
```

REQUEST/RESPONSE PAIR 2

REQUEST:

```
{
  "query": {
    "description": "The search query for news",
    "type": "string"
  },
  "required": [
    "query"
  ],
  "type": "object"
},
{
  "type": "function"
}
]
```

RESPONSE:

```
{
  "created": 1780049900,
  "id": "chatcmpl-abd6170b-lee6-9153-90c9-cf53c2579e63",
  "model": "qwen3-max",
  "object": "chat.completion",
  "usage": {
    "completion_tokens": 215,
    "prompt_tokens": 2437,
    "prompt_tokens_details": {
      "cached_tokens": 0
    },
    "total_tokens": 2652
  }
}
```

输入消息...

🗣️

Send

Agent--Phase 6: Adding Search and Download

Enable AI to search the web and download resources

Functions:

| Function | Description |
|----------------|------------------------------------|
| web_search | Web search |
| news_search | News search |
| download_image | Download images to files directory |

Use Cases:

User: Help me search for Python tutorials

Agent: [calls web_search] → Returns search results

User: Help me download a picture of the Forbidden City

Agent: [calls download_image] → Image saved to files/

Agent--Phase 6: Adding Search and Download

Agent Chat

Clear

AGENT

→ 发送请求到 LLM

← LLM 返回, 准备调用工具

X 调用 web_search... 失败

→ 发送请求到 LLM

← LLM 返回, 准备调用工具

✓ 调用 write_file... 成功

→ 发送请求到 LLM

← LLM 返回最终回复

Behold, FQM! 🏰🌟

I've built you a ****Forbidden City mini-exhibit**** — right in your browser!

📁 File saved as: 'files/forbidden_city.html'

🖼️ It displays a stunning ****aerial view of the Forbidden City**** (courtesy of Wikimedia Commons, public domain — so even the Emperor would approve!).

🎨 Styled with imperial reds and warm parchment tones... because why settle for boring when you can have ***majesty***?

Just open 'files/forbidden_city.html' in your browser, and boom — you're virtually strolling through 600 years of history! 🏰

Want me to add:

- A "Random Fact About the Forbidden City" button?
- Background music (imperial gongs, anyone?)?
- Or turn it into a full Beijing travel guide?

The throne room is yours! 😊

MESSAGE DETAILS

```
{
  "completion_tokens": 799,
  "prompt_tokens": 2808,
  "prompt_tokens_details": {
    "cached_tokens": 2624
  },
  "total_tokens": 3607
}
```

REQUEST/RESPONSE PAIR 3

REQUEST:

```
{
  "query": {
    "description": "The search query for news",
    "type": "string"
  },
  "required": [
    "query"
  ],
  "type": "object"
},
{
  "type": "function"
}
]
```

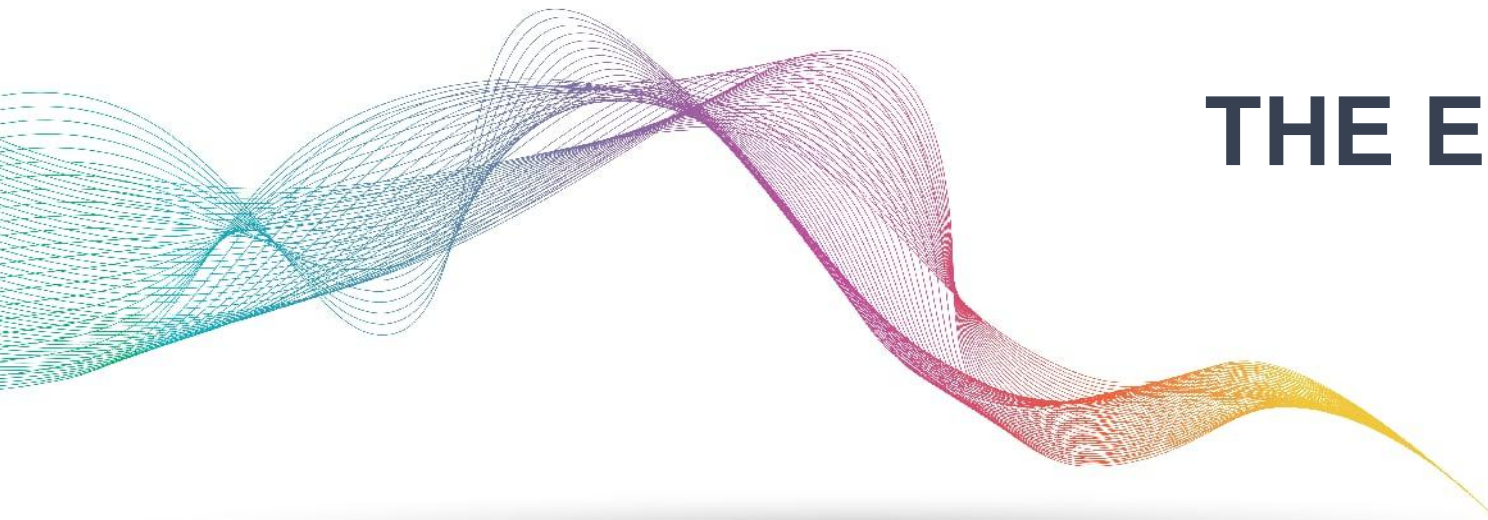
RESPONSE:

```
{
  "created": 1780050142,
  "id": "chatcmp1-c6d833f6-ef46-95d9-a242-9cff7a02840d",
  "model": "qwen3-max",
  "object": "chat.completion",
  "usage": {
    "completion_tokens": 184,
    "prompt_tokens": 3186,
    "prompt_tokens_details": {
      "cached_tokens": 0
    },
    "total_tokens": 3370
  }
}
```

输入消息...

🗣️

Send



THE END